

Jay Desai

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SUMMARY

PhD candidate in theoretical particle physics with **four** research papers (three published in Physical Review D, fourth as an arXiv preprint), **70+** citations, **eight** conference talks, collaborated in multi-institutional research teams and invited seminars at Brookhaven National Lab and CERN. Specializing in large-scale numerical computation, symbolic manipulation, and multi-parameter optimization applied to high-dimensional scientific datasets. Interested in foundation models for scientific data, building ML surrogate and physics-informed models from large-scale simulation data.

EDUCATION

- **Stony Brook University** 2020 – Present
Ph.D. in Physics (High Energy Theory and Phenomenology)
- **Stony Brook University** 2018 – 2020
M.A. in Physics
- **Indian Institute of Technology Roorkee** 2014 – 2018
B.Tech., Electronics and Communication Engineering, Minor in Physics

RESEARCH EXPERIENCE

- **Graduate Research Assistant, C.N. Yang Institute for Theoretical Physics** 2020 – Present
Stony Brook University
 - Co-developed an end-to-end analysis pipeline: derived model predictions symbolically in Mathematica, fed them into MadGraph for Monte Carlo event generation at scale (**1M+** events), and processed simulation output for statistical fits and visualization in Python and C++.
 - Aggregated and harmonized **85+** measurements from heterogeneous experimental sources (LEP/SLD electroweak precision data, ATLAS and CMS experiments) into a unified statistical analysis, accounting for correlations among inputs.
 - Performed global likelihood fits over an **8-dimensional** parameter space via chi-squared minimization; profiled and marginalized nuisance directions, extracted confidence regions, and identified and resolved parameter degeneracies via reparametrization.
 - Constructed dimension-eight basis of **175** operators for universal effective field theories filtered from **44,807**. Cross-checked using group theory and representation theory techniques to compute group invariants.
- **Research Assistant, Center for Nuclear Science** 2018 – 2020
Stony Brook University
 - Experience with large-scale simulation packages generating **1M+** events and C++ based data analysis framework ROOT, executed via HTCondor job submission on the Brookhaven National Lab computing cluster.
 - Applied statistical reweighting techniques to estimate projected uncertainty reduction of $\sim$ **20%** for the proposed electron-ion collider.

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, Scikit-learn, Matplotlib, Pandas, Jupyter, statistical inference, high-dimensional modeling, model validation, data analysis
- **Programming:** Python, C, C++, Java, SQL
- **Mathematical Methods:** Probability theory, stochastic processes, hypothesis testing, multi-parameter regression
- **Scientific Computing:** SciPy, NumPy, Mathematica, MATLAB, ROOT, IBM Qiskit
- **High-Performance Computing:** HTCondor, Linux cluster computing
- **Simulation & Modeling:** Monte Carlo Methods: MadGraph
- **Tools:** Git, LaTeX, Claude Code

CONFERENCE/INVITED TALKS

- **Brookhaven National Lab (BNL) Seminar (invited), Brookhaven, December 11th 2025**
“Universal Standard Model Effective Field Theory beyond dimension-six”
- **Higgs and Effective Field Theory (HEFT), CERN, June 3rd 2025**
“Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
- **Large Hadron Collider (LHC) EFT Working Group meeting, (invited), April 28th 2025**
“Drell-Yan production in universal theories beyond dimension-six SMEFT”
- **American Physical Society (APS) DPF-Pheno 2024, University of Pittsburgh, May 13th 2024**
“Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
- **American Physical Society (APS) April Meeting, 2020**
“Strange quark polarized distribution function using SIDIS at EIC”

PROJECTS

- **Predicting Award Nominations for Sci-Fi/Fantasy Novels** Erdős Institute Data Science Boot Camp
GitHub
 - Built learning-to-rank models (LightGBM/XGBoost LambdaRank, grouped by year) in a team of five to predict Hugo and Locus Award finalists from 50+ years of book data with severe class imbalance ($\sim 1.3\%$ positives), achieving a $\sim 1.7\times$ improvement in F_1 over logistic regression and prior-nominee baselines.
 - Engineered features from multi-source data (ISFDB SQL dump, Hardcover API, SFADB): year-cohort z-scored author/publisher prestige metrics, cosine similarity to past nominees in tag space, and 384-dim synopsis embeddings (all-MiniLM-L6-v2 sentence transformer), with custom sklearn transformers to prevent temporal leakage.
 - Evaluated via nested walk-forward time-series cross-validation (train 1971–2014, test 2015–2025) using ranking metrics (Precision@k, NDCG@k) alongside F_1 score; diagnosed and fixed a data leakage bug in cumulative author-history features.
- **Forecasting 45-Day Forward Stock Beta**
Quant assessment mini-project
 - Built an end-to-end ML pipeline to predict 45-day forward-looking beta relative to SPY from daily price/volume panel data; engineered 15 features across multi-horizon trailing betas, correlation/volatility decomposition, beta instability, market-state, and liquidity/momentum groups, with universe cleaning to remove leveraged/inverse ETFs contaminating the loss.
 - Combined ridge regression and LightGBM into a 50/50 ensemble evaluated via purged walk-forward cross-validation (45-day purge to prevent label-overlap leakage); on the held-out test period the ensemble reduced winsorized MSE by $\sim 2.6\%$ over a fitted Blume-shrinkage baseline.
- **Multilayer Perceptron from Scratch** CSE 353 Machine Learning
Stony Brook University
 - Implemented a multilayer perceptron in PyTorch with manual forward/backward passes and gradient derivations; trained and validated on classification datasets, evaluating generalization across decision boundaries.

PUBLICATIONS

All papers can be found on my [INSPIRE](#) and [Google Scholar](#) profile. Authors ordered alphabetically.

- “USMEFT as a tool for discovery of universal new physics at high luminosity LHC”
[arXiv:2605.15262 \[hep-ph\]](#)
Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martines
- “Drell-Yan production in universal theories beyond dimension-six SMEFT”
[arXiv:2503.19962 \[hep-ph\]](#), *Phys. Rev. D* **112**, 013009 (2025)
Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz
- “Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
[arXiv:2404.03720 \[hep-ph\]](#), *Phys. Rev. D* **110** (2024) 033003
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia
- “Impact of dimension-eight SMEFT operators in the EWPO and Triple Gauge Couplings analysis in Universal SMEFT”
[arXiv:2304.03305 \[hep-ph\]](#), *Phys. Rev. D* **107**, 115013 (2023)
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz

AWARDS & ACHIEVEMENTS

- Gerald Brown Prize (Research Travel Award), Stony Brook University
- Nationwide Education and Scholarship Test (Ranked **13th** nationally)
- Joint Entrance Examination (JEE) Advanced (Ranked **971** nationally)

LEADERSHIP & TEACHING

- Instructor for PHY 121, PHY 122, PHY 133, PHY 134.
- Teaching Assistant for PHY 301, PHY 308, PHY 503.
- Journal Referee for the Journal of High Energy Physics.